

Tautology

Compound statement

↳ true

for all truth values
of its component
statements

Contradiction

Compound statement

↳ false

for all truth values
of its component
statements

premises: $p_1, p_2, \dots, p_n$

$p_1 \wedge p_2 \wedge p_3 \wedge \dots \wedge p_n \rightarrow q$ — true

↓
valid argument.

tautology

E.g. 1.

$$(p \vee q) \vee [\neg p \wedge \neg q]$$

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg p \wedge \neg q$	$(p \vee q) \vee (\neg p \wedge \neg q)$
0	0	1	1	0	1	1
0	1	1	0	1	0	1
1	0	0	1	1	0	1
1	1	0	0	1	0	1

→ tautology.

E.g. 2.

$$(p \vee q) \wedge [\neg p \wedge \neg q]$$

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg p \wedge \neg q$	$(p \vee q) \wedge (\neg p \wedge \neg q)$
0	0	1	1	0	1	0
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	1	0	0

→ Contradiction.

Logical equivalence.

Two statements S_1 & S_2 are said to be logically equivalent if when the statement S_1 is true (or respectively false) if and only if statement S_2 is true (or respectively false).

$$S_1 \Leftrightarrow S_2.$$

p	q	$\neg p$	$p \rightarrow q$	$\neg p \vee q$
0	0	1	1	1
0	1	1	1	1
1	0	0	0	0
1	1	0	1	1

$$p \rightarrow q \Leftrightarrow \neg p \vee q$$

H.W. Can you establish a valid argument from this?

E.g. 2

$$(p \rightarrow q) \wedge (q \rightarrow p) \rightarrow p \leftrightarrow q.$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$x = (p \rightarrow q) \wedge (q \rightarrow p)$	$y = p \leftrightarrow q$	$x \rightarrow y$
0	0	1	1	1	1	1
0	1	1	0	0	0	1
1	0	0	1	0	0	1
1	1	1	1	1	1	1

tauto logy.

logically
equivalent

$$p \vee q \Leftrightarrow (p \vee q) \wedge \neg(p \wedge q).$$

Algebraic Laws of Logic.

Law of double negation.

$$(1) \neg\neg p \Leftrightarrow p$$

$$(2) \left. \begin{aligned} \neg(p \vee q) &\Leftrightarrow \neg p \wedge \neg q \\ \neg(p \wedge q) &\Leftrightarrow \neg p \vee \neg q \end{aligned} \right\} \text{De Morgan's Law.}$$

$$(3) \left. \begin{aligned} p \vee q &\Leftrightarrow q \vee p \\ p \wedge q &\Leftrightarrow q \wedge p \end{aligned} \right\} \text{Commutative Law.}$$

$$(4) \left. \begin{aligned} p \vee (q \vee r) &\Leftrightarrow (p \vee q) \vee r \\ p \wedge (q \wedge r) &\Leftrightarrow (p \wedge q) \wedge r \end{aligned} \right\} \text{Associative Law.}$$

$$(5) \left. \begin{aligned} p \vee (q \wedge r) &\Leftrightarrow (p \vee q) \wedge (p \vee r) \\ p \wedge (q \vee r) &\Leftrightarrow (p \wedge q) \vee (p \wedge r) \end{aligned} \right\} \text{Distributive Law.}$$

$$(6) \left. \begin{aligned} p \vee p &\Leftrightarrow p \\ p \wedge p &\Leftrightarrow p \end{aligned} \right\} \text{Idempotent Law.}$$

$$(7) \left. \begin{aligned} p \vee F_0 &\Leftrightarrow p \\ p \wedge T_0 &\Leftrightarrow p \end{aligned} \right\} \text{Identity Laws.}$$

$$\textcircled{8} \quad \begin{array}{l} p \vee \neg p \Leftrightarrow T_0 \\ p \wedge \neg p \Leftrightarrow F_0 \end{array} \quad \left. \vphantom{\begin{array}{l} p \vee \neg p \Leftrightarrow T_0 \\ p \wedge \neg p \Leftrightarrow F_0 \end{array}} \right\} \text{Inverse Law}$$

$$\textcircled{9} \quad \begin{array}{l} p \vee T_0 \Leftrightarrow T_0 \\ p \wedge F_0 \Leftrightarrow F_0 \end{array} \quad \left. \vphantom{\begin{array}{l} p \vee T_0 \Leftrightarrow T_0 \\ p \wedge F_0 \Leftrightarrow F_0 \end{array}} \right\} \text{Domination Law}$$

$$\textcircled{10} \quad \begin{array}{l} p \vee (p \wedge q) \Leftrightarrow p \\ p \wedge (p \vee q) \Leftrightarrow p \end{array} \quad \left. \vphantom{\begin{array}{l} p \vee (p \wedge q) \Leftrightarrow p \\ p \wedge (p \vee q) \Leftrightarrow p \end{array}} \right\} \text{Absorption Law.}$$

Duality of propositional statements.

$S \leftarrow$ statement

If S contains no other connectives other than $\vee$ & $\wedge$ then the dual of S , denoted by S^d is obtained by replacing each occurrence of $\wedge$ & $\vee$ with $\vee$ & $\wedge$ respectively.

Each T_0 is replaced by F_0
 $\triangleright F_0$ is " " " T_0

E.g. 4.

$$S: (p \wedge \neg q) \vee (r \wedge T_0)$$

$$S^d: (p \vee \neg q) \wedge (r \vee F_0).$$

Principle of duality.

Let us say S & t are two statements
s.t. $S \Leftrightarrow t$

then

$$S^d \Leftrightarrow t^d$$

In all the laws & expressions $\rightarrow$
primitives can be as well substituted by
longer compound statements.

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$[\overline{p \wedge (p \rightarrow q)}] \rightarrow q$
0	0	1	0	1
0	1	1	0	1
1	0	0	0	1
1	1	1	1	1

$$[(\sigma \rightarrow s) \wedge [(\sigma \rightarrow s) \rightarrow (\neg t \vee u)]] \rightarrow (\neg t \vee u).$$

is also a tautology

p: $\sigma \rightarrow s$

q: $\neg t \vee u$

Using Laws for logical deduction/
Simplification.

$$\begin{aligned}
 & (p \vee q) \wedge \neg(\neg p \wedge q) \\
 \Leftrightarrow & (p \vee q) \wedge (\neg \neg p \vee \neg q) \\
 \Leftrightarrow & (p \vee q) \wedge (p \vee \neg q) \\
 \Leftrightarrow & p \vee (q \wedge \neg q) \\
 \Leftrightarrow & p \vee F_0 \\
 \Leftrightarrow & p
 \end{aligned}$$

De Morgan's law.

Double negation.

Distributive law.

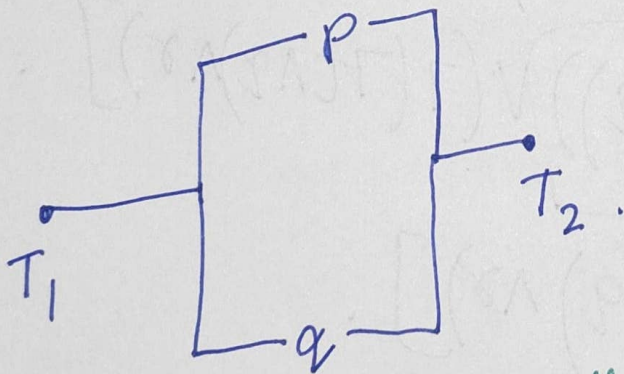
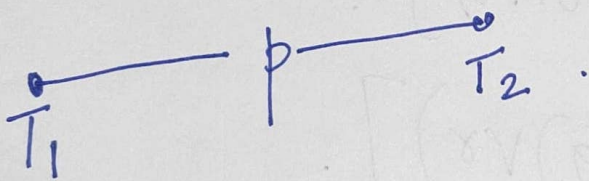
Inverse Law

Identity law.

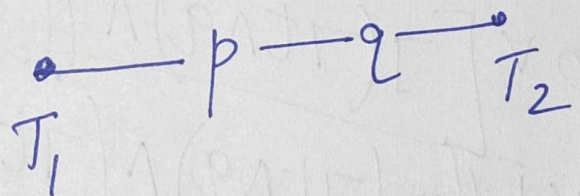
Switching network

Logical statements $\rightarrow$ Switching network.

Terminals.
Switches $\rightarrow$ closed $\Rightarrow 1$
open $\Rightarrow 0$.



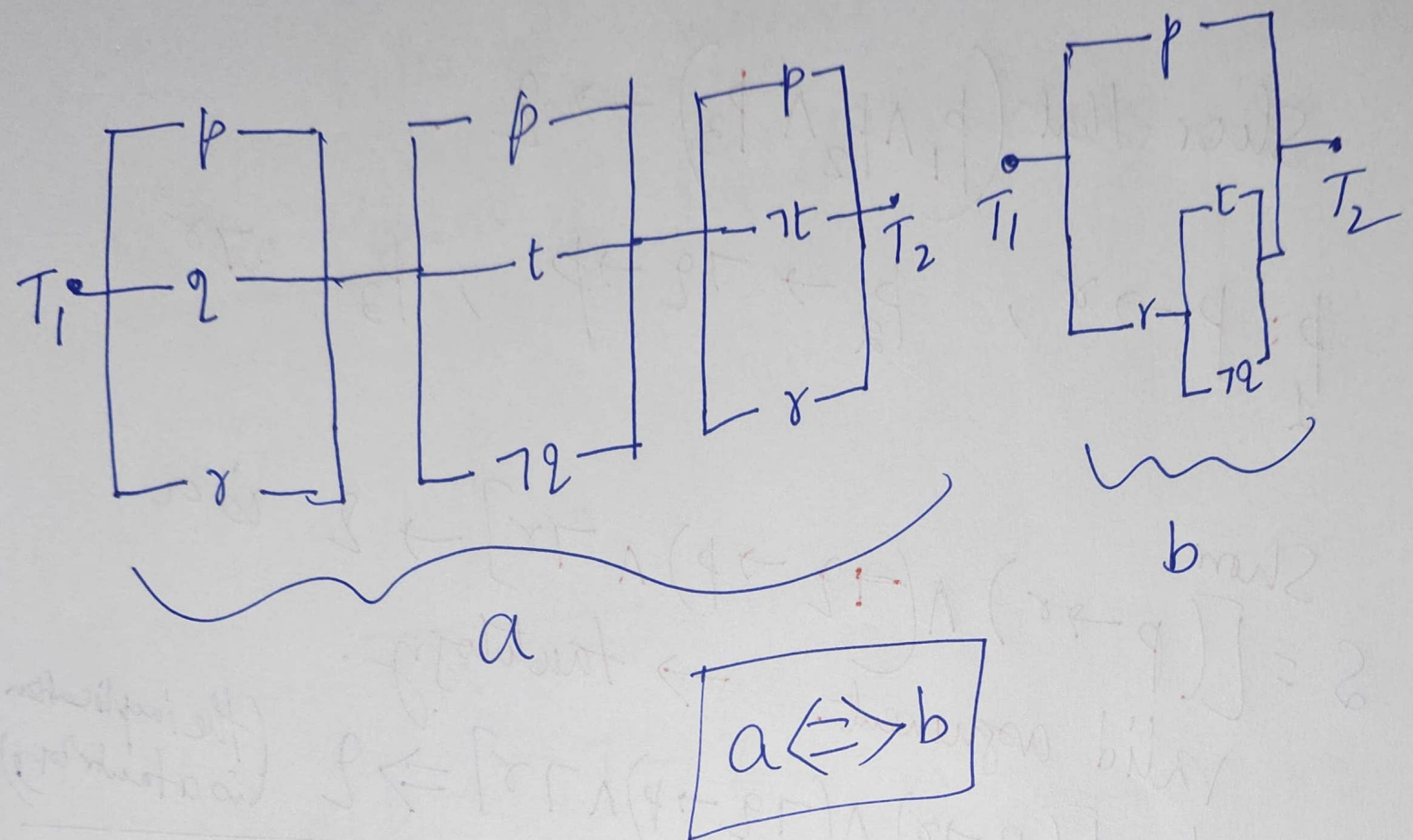
$(p \vee q) \rightarrow$ parallel



$(p \wedge q) \rightarrow$ series.

E.g.

- $$(p \vee q \vee r) \wedge (p \vee t \vee \neg q) \wedge (p \vee \neg t \vee r)$$
- $$\Leftrightarrow p \vee [(q \vee r) \wedge (t \vee \neg q) \wedge (\neg t \vee r)] \rightarrow \text{Distributive.}$$
- $$\Leftrightarrow p \vee [(q \vee r) \wedge (\neg t \vee r) \wedge (t \vee \neg q)] \rightarrow \text{Commutative.}$$
- $$\Leftrightarrow p \vee [((q \wedge \neg t) \vee r) \wedge (t \vee \neg q)] \rightarrow \text{Distributive.}$$
- $$\Leftrightarrow p \vee [(q \wedge \neg t) \vee r \wedge (\neg t \vee \neg q)]$$
- $$\Leftrightarrow p \vee [(q \wedge \neg t) \vee r \wedge (\neg t \wedge q)]$$
- $$\Leftrightarrow p \vee [(q \wedge \neg t) \vee r \wedge (\neg t \wedge q)]$$
- $$\Leftrightarrow p \vee [\neg(\neg t \wedge q) \wedge ((\neg t \wedge q) \vee r)]$$
- $$\Leftrightarrow p \vee [(\neg(\neg t \wedge q) \wedge (\neg t \wedge q)) \vee (\neg(\neg t \wedge q) \wedge r)]$$
- $$\Leftrightarrow p \vee [\bar{r} \vee \neg(\neg t \wedge q) \wedge r]$$
- $$\Leftrightarrow p \vee [\neg(\neg t \wedge q) \wedge r]$$
- $$\Leftrightarrow p \vee [r \wedge \neg(\neg t \wedge q)]$$
- $$\Leftrightarrow p \vee [r \wedge (t \vee \neg q)]$$



E.g. $p, q, r \rightarrow$ primitive statements.

p : Roger studies.

q : Roger plays tennis.

r : Roger passes discrete maths.

p_1, p_2, p_3 :

p_1 : If Roger studies then he will pass discrete math.

p_2 : If Roger doesn't play tennis, then he will study.

p_3 : Roger failed in discrete math.

Show that $(p_1 \wedge p_2 \wedge p_3) \rightarrow 2$.

$p_1: p \rightarrow r$, $p_2: \neg q \rightarrow p$, $p_3: \neg r$

Show $S = [(p \rightarrow r) \wedge (\neg q \rightarrow p) \wedge \neg r] \rightarrow 2$ is a valid argument. $\rightarrow$ tautology.

$\hookrightarrow [(p \rightarrow r) \wedge (\neg q \rightarrow p) \wedge \neg r] \Rightarrow 2$ (the implication is a tautology)

p	q	r	$p \rightarrow q$	$\neg q \rightarrow p$	$\neg r$	S
0	0	0	1	0	1	1
0	0	1	1	0	0	1
0	1	0	1	1	1	1
0	1	1	1	1	0	1
1	0	0	0	1	1	1
1	0	1	1	1	0	1
1	1	0	0	1	1	1
1	1	1	1	1	0	1

If p & q are any arbitrary statements and $p \rightarrow q$ is a tautology $\Rightarrow$ p logically implies q ($p \Rightarrow q$)

Rules of inference.

Modus Ponens (Rule of Detachment)

$$\frac{p \quad p \rightarrow q}{\therefore q}$$

$$[p \wedge (p \rightarrow q)] \rightarrow q$$

p	q	p → q	p ∧ (p → q)	[p ∧ (p → q)] → q
0	0	1	0	1
0	1	1	0	1
1	0	0	0	1
1	1	1	1	1

$$[p \wedge (p \rightarrow q)] \Rightarrow q$$

- ① If Alice vacations in Paris then she will have to win a scholarship
- ② Alice is vacationing in Paris
- ③ Therefore Alice has won the scholarship.

Law of Syllogisms.

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r. \end{array}$$

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \Rightarrow (p \rightarrow r)$$

-
- ① Rita is baking a cake.
- ② If Rita is baking a cake, then she is not practicing her flute.
- ③ If Rita is not practicing her flute then her father will not pay for her car insurance.
- ④ Therefore Rita's father will not pay her car insurance.

$$\begin{array}{l} p \\ p \rightarrow \neg q \\ \neg q \rightarrow \neg r \\ \hline \therefore \neg r \end{array} \quad \left. \vphantom{\begin{array}{l} p \\ p \rightarrow \neg q \\ \neg q \rightarrow \neg r \end{array}} \right\} \begin{array}{l} \text{Syllogism.} \\ \rightarrow \end{array}$$

$$\begin{array}{l} p \rightarrow \neg r \\ \downarrow \\ p \\ p \rightarrow \neg r \\ \hline \therefore \neg r \end{array} \quad \left. \vphantom{\begin{array}{l} p \rightarrow \neg r \\ p \\ p \rightarrow \neg r \end{array}} \right\} \begin{array}{l} \text{Modus} \\ \text{Ponens} \end{array}$$

Rule of conjunction.

$$\begin{array}{c} p \\ q \\ \hline \therefore p \wedge q \end{array}$$

$$(p \wedge q) \Rightarrow (p \wedge q).$$

Rule of disjunctive Syllogism.

$$\begin{array}{c} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

$$[(p \vee q) \wedge \neg p] \Rightarrow q.$$

Rule of Contradiction.

$$\frac{\neg p \rightarrow F_0}{\therefore p}$$

$$(\neg p \rightarrow F_0) \Rightarrow p.$$

Modus Ponens (2)

Modus Ponens (1)

① p
 $p \rightarrow \neg q$

 $\neg q$

$\neg q$
 $\neg q \rightarrow \neg r$

 $\neg r$

(2)

Modus Tollens

$p \rightarrow q$
 $\neg q$

 $\therefore \neg p$

E.g. Show that the following argument is valid.

$p \rightarrow r$ $r \rightarrow s$ $t \vee \neg s$ $\neg t \vee u$ $\neg u$ $\therefore \neg p$	$\rightarrow$ $\cdot$ $\rightarrow$ $\rightarrow$	$p \rightarrow s$ (syl) $\neg s \vee t \Leftrightarrow s \rightarrow t$ $p \rightarrow t$ (syl) $t \rightarrow u$ $p \rightarrow u$ (syl) $\neg u$ $\neg p$
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Modus Tollens

Rule of Conjunctive simplification.

$$\frac{p \wedge q}{\therefore p}$$

$$(p \wedge q) \Rightarrow p.$$

Rule of Disjunctive amplification

$$\frac{p}{\therefore p \vee q}$$

$$p \Rightarrow p \vee q.$$

Rule of conditional proof.

$$\frac{\begin{array}{l} p \wedge q \\ p \rightarrow (q \rightarrow r) \end{array}}{\therefore r} \quad [(p \wedge q) \wedge p \rightarrow (q \rightarrow r)] \Rightarrow r.$$

Rule by cases.

$$\frac{\begin{array}{l} p \rightarrow r \\ q \rightarrow r \end{array}}{\therefore (p \vee q) \rightarrow r}$$

Rule of Constructive Dilemma.

$$\begin{array}{l} p \rightarrow q \\ r \rightarrow s \\ p \vee r \\ \hline \therefore q \vee s \end{array}$$

Rule of Destructive Dilemma.

$$\begin{array}{l} p \rightarrow q \\ r \rightarrow s \\ \neg q \vee \neg s \\ \hline \therefore \neg p \vee \neg r \end{array}$$

E.g.

Show that

$$\begin{array}{c} p \rightarrow r \\ \neg p \rightarrow q \\ q \rightarrow s \\ \hline \therefore \neg r \rightarrow s \end{array}$$

is a valid argument.

Catch is:

$$p \rightarrow r \Leftrightarrow \left. \begin{array}{l} \neg r \rightarrow \neg p \\ \neg p \rightarrow q \end{array} \right\}$$

$$\begin{array}{l} \therefore \neg r \rightarrow q \quad \leftarrow \text{syll} \\ q \rightarrow s \quad \leftarrow \text{syll} \\ \hline \end{array}$$

H.W.

$$\begin{array}{c} (\neg p \vee \neg q) \rightarrow (r \wedge s) \\ r \rightarrow t \\ \neg t \\ \hline \therefore p \end{array}$$

Whether the following are valid arguments?

$$\frac{\neg r \rightarrow s}{p}$$

$$\begin{array}{c} p \vee q \\ q \rightarrow (r \rightarrow s) \\ t \rightarrow r \\ \hline \end{array}$$

$$\therefore \neg s \rightarrow \neg t$$